

Omar Elmasaoudi

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SUMMARY

Software developer with experience in full-stack development, ML/NLP, and System Design. Skilled in building scalable web applications, implementing different web architectures, and applying modern LLM models using techniques like fine-tuning, RLHF, and prompt engineering for diverse applications.

EDUCATION

Concordia University

Bachelor of Engineering in Software Engineering

Montreal, QC

September 2022 – May 2026

- **Core SE Courses:** Data Structures & Algorithms, Software Architecture, Software Processes & Lifecycle
- **Programming:** Object-Oriented Programming (Java), Web Development (PHP, JavaScript), Functional Programming (Erlang, Clojure)
- **Database Management:** Database Design, SQL, Data Modeling

WORK EXPERIENCE

Air Ocean Maroc

Software Developer Intern

Rabat, Morocco

April 2024 – August 2024

- Collaborated with DevOps engineers to deploy a Medivac software system to the company's network, ensuring seamless integration and scalability.
- Worked closely with medics and operations personnel to verify that all requirements were met, addressing real-time scheduling, supply tracking, and secure access to patient and personnel data.
- Enhanced flight and ground Medivac operations efficiency by 40% through the implementation of key features such as real-time scheduling and supply tracking.

PROJECTS

Stock Performance and Analysis Tool | *Python, NextJs, Typescript, MongoDB*

November 2024 – Present

- Built a portfolio management tool using **NLP** to analyze financial articles with **85% prediction accuracy**, providing actionable insights for investment decisions.
- Engineered a financial analysis pipeline using **pre-trained language models** with **transfer learning** to analyze sentiment across multiple financial sources, achieving a **78% sentiment classification accuracy**. Implemented a **Neo4j graph database** to visualize dynamic company relationships and market influences.
- Developed a responsive **Next.js** frontend with **TypeScript** backend and Python **web scraper** that processes **10,000+** financial articles weekly.

Managed Medivac Application | *React, Typescript, PostgreSQL, Docker*

February 2024 – April 2024

- Created a central platform for medivac operations, reducing response times by **35%** and improving resource allocation by **40%**.
- Built responsive UI with **React**, **Vite**, and **Chakra UI**, achieving **95%** adoption rate among staff.
- Architected a scalable **microservices** infrastructure secured via custom **TypeScript** ID provider, NGINX reverse proxy, and **JWT**-based authentication, handling **500+** concurrent user sessions without performance degradation.
- Automated deployment with **Docker** CI/CD pipeline, reducing update times from hours to **15 minutes**.

Operations Manager Application | *JavaScript, Express.js, React, MySQL*

May 2023 – August 2023

- Developed inventory management system that reduced discrepancies by **65%** and improved fulfillment by **30%**.
- Built **JavaScript**, **Express.js**, and **React** stack with RESTful API using **JWT** authentication and validation middleware.
- Implemented automated stock alerts preventing **15+ critical stock-outs** monthly and saving **8 hours** weekly.
- Secured medical data with role-based authorization and **AES-256 encryption**, maintaining full compliance.
- Deployed to **AWS EC2** with auto-scaling for **10+** employees, ensuring **99.9%** uptime during server migration.
- Optimized **MySQL** database schema reducing query times by **45%** through indexing and stored procedures.

CERTIFICATIONS

DeepLearning.AI Machine Learning Specialization

May 2024 – July 2024

- Learned foundational concepts of supervised learning, including **linear regression**, **logistic regression**, **neural networks**, and **decision trees**. Gained expertise in unsupervised learning techniques like **clustering** and **anomaly detection** to analyze and model unlabeled data.
- Applied advanced techniques to build **recommender systems**, leveraging **collaborative filtering** and **content-based methods**. Explored practical applications of deep learning with projects on **CNNs**, **optimization algorithms**, and **reinforcement learning**.

DeepLearning.AI Natural Language Processing (NLP) Specialization

August 2024 – November 2024

- Mastered techniques including **logistic regression**, **naïve Bayes**, and **word vectors** to implement sentiment analysis, complete analogies, and translate words. Used **dynamic programming**, **hidden Markov models**, and **word embeddings** to build autocorrect, autocomplete, and identify part-of-speech tags for words.
- Developed advanced models like **recurrent neural networks (RNNs)**, **LSTMs**, **GRUs**, and **Siamese networks** for sentiment analysis, text generation, and named entity recognition. Applied **encoder-decoder models**, **causal**, and **self-attention mechanisms** for tasks like machine translation, text summarization, and question answering.

DeepLearning.AI Convolutional Neural Networks (CNN) Course

July 2024 – August 2024

- Built **convolutional neural networks (CNNs)**, including advanced models like **Residual Networks (ResNets)**, **MobileNet**, and **EfficientNet**. Applied CNNs to tasks such as **object detection**, **recognition**, and **landmark detection**, leveraging techniques like **YOLO** and **U-Net** for **semantic segmentation** and **bounding box predictions**.
- Explored advanced applications of CNNs, including **face recognition** using **one-shot learning**, **Siamese networks**, and **triplet loss**. Developed **neural style transfer** models to generate art and applied convolutional techniques to **image**, **video**, and **3D data** through **transfer learning** and **data augmentation**.

DeepLearning.AI TensorFlow Developer Professional Certificate

August 2024 – Present

- Learned best practices for using **TensorFlow** to train neural networks for computer vision applications. Handled real-world image data and implemented strategies to prevent overfitting, including **data augmentation** and **dropout**.
- Built natural language processing systems using **TensorFlow**, applying **RNNs**, **GRUs**, and **LSTMs** to train models on text repositories for tasks such as text generation and sentiment analysis.

Docker and Kubernetes Complete Guide | Udemy

February 2024 - March 2024

- Mastered containerization concepts with **Docker**, including building, deploying, and managing containerized applications, and leveraging Docker Compose for multi-container environments.
- Learned orchestration with **Kubernetes**, including setting up clusters, managing deployments, scaling applications, and configuring services using best practices for high availability and fault tolerance.

HACKATHONS

@Hack CTF Cybersecurity Competition

March 3, 2024

- Solved network analysis, cryptography, and reverse engineering challenges in a 24-hour cybersecurity competition. Applied **penetration testing** tools and techniques to identify vulnerabilities and secure compromised systems.

SKILLS

Programming Languages: Java, Python, Typescript

Frameworks: TensorFlow, Numpy, Pandas, NodeJs, Nextjs, Tailwind CSS

Databases: Postgres, Mysql, Neo4j, MongoDB, Redis

Developer Tools: Git, AWS, Docker, Vercel

Languages: English (Native), French (Fluent), Arabic (intermediate)